Advanced AI Skills

Verification, Governance and the Real World

Real Minds AI

Tracy Anthony and Dr Dennis Wollersheim

Why "Advanced"?

The morning exercises used clean data. Real life is messy.
In the real world, the data you feed AI is:

- **Incomplete** – fields missing, logs partial, records half-finished
- **Contradictory** – two people remember the same event differently
- **Ambiguous** – "soon", "when convenient", "where appropriate"
- **Risky** – one wrong word in an email can breach privacy or over-promise

This session teaches you to use AI safely when the inputs are imperfect

> *The question is not "can AI write this?" but "should I send what AI wrote?"*

The AI Sandwich

The workflow that keeps you safe

Step 1: System A outputs (raw notes, exports, logs, emails)
These are messy, partial, sometimes contradictory

Step 2: AI transforms (triage, structure, draft comms)
AI imposes structure and generates drafts – fast but unchecked

Step 3: Human verifies (facts, privacy, policy, approvals)
You check every claim, every name, every commitment

Step 4: System B inputs (incident record, complaint register, final comms)
Clean, verified, auditable output goes into your systems

> *AI sits in the middle. **You are the bread.** Nothing gets through without your verification.*

Advanced Exercise Overview

#	Exercise	What you will do	Key skill
A1	**Incident Pack Assembly**	Reconcile 2 conflicting accounts into one record	Verification
A2	**Fix the Problematic Email**	Rewrite a privacy-breaching draft	Governance
A3	**Attendance Triage Under Uncertainty**	Build contact plan from messy data	Uncertainty
A4	**Policy Alignment Change Log**	Compare old procedure to new requirements	Compliance
A5	**Messy Minutes to Action Register**	Extract clean actions from vague notes	Structure

Your data pack

The **Messy Mini-Pack** is intentionally broken: duplicate rows, conflicting times, missing fields, privacy traps, and unclear ownership. This is realistic.

Exercise A1

Incident Pack Assembly

Verification and Cross-Checking

The theory: when sources disagree

- In any incident, different people observe different things at different times
- Two staff accounts of the same event **will** contain discrepancies (times, sequence, severity)
- Your job is not to pick one account – it is to **reconcile** them into a single factual record

The Known / Unconfirmed / Unknown framework

Category	Definition	Example
Known	Confirmed by 2+ sources or physical evidence	"S1188 had a graze – confirmed by first aid log"
Unconfirmed	Mentioned by one source, not contradicted	"Bystander called out a name – one account only"
Unknown	Key fact that no source provides	"Exact start time of the altercation"

> *AI can build this framework from messy inputs. **You** decide what moves from "Unconfirmed" to "Known".*

Exercise A1: Incident Pack Assembly

Instructions (20 minutes)

Open **5 documents** from your Messy Mini-Pack: Artifacts 02-06 (yard duty account, student services note, first aid log, parent contact log, CCTV ticket)

Paste your context card, then **all five documents**, then add:

> *Using these five documents about the same incident, please:*
> *1. Create a Known / Unconfirmed / Unknown fact list*
> *2. Produce a SafeTrack incident entry with structured fields (Date, TimeRange, Location, Category, PeopleInvolved, ImmediateActions, Injuries, ParentContact, Evidence, FollowUpOwner, ReviewDate)*
> *3. List what must be verified by a human before finalising*
>
> *Do not invent facts. Clearly mark all conflicts between sources.*

What to look for

- Does the AI catch the **time discrepancy** between the two accounts?
- Does it flag the CCTV "no signal" gap and the "punch" claim?

Exercise A1: What You Learned

Key contradictions in the data

- **Time:** Yard duty says "approx 13:05" arrival, Student Services says students arrived "approx 13:12" – times are estimates
- **Physical contact:** Yard duty is uncertain about a punch; both students told Student Services there was no punching
- **First aid log:** Two entries for S1188 at different times (13:16 and 13:18) – possible duplicate or two visits
- **CCTV:** Dashboard showed "NO SIGNAL" during the exact window (13:10-13:25) – footage may not exist
- **Parent contact:** S1048 contact logged at 13:25 (attempt) and 13:32 (success) – Student Services notes "13:25" but may be the attempt time

The principle

> *AI organises conflicting information. **You** verify which facts are confirmed. Never let AI resolve contradictions – that is a human judgment call.*

Exercise A2

Fix the Problematic Email

Spot the Problems

This draft email contains at least 6 governance failures

Read the **Problematic Draft Email** (Artifact 07 in your Messy Mini-Pack):

Problem	Category
Implies the other student's identity	**Privacy breach**
Says student "has a history of behaviour issues"	**Privacy breach**
Promises "suspending them for 5 days"	**Over-promise** (not yet decided)
Claims "CCTV clearly shows this"	**False claim** (CCTV had no signal)
Promises "this will never happen again"	**Over-promise**
Asks for "screenshots of the group chat"	**Evidence risk** (other students' privacy)

> *Every one of these could trigger a formal complaint or privacy investigation.*

Exercise A2: Fix the Problematic Email

Instructions (15 minutes)

Paste your context card, then **Artifact 07** (Problematic Draft Email), then add:

> *This draft email must not be sent as-is. Rewrite it so that it: does not identify any other student, does not promise specific outcomes, does not claim unverified evidence, includes next steps with a timeframe and contact point, and uses a calm, empathetic tone.*
>
> *Provide three versions: A. Phone call script (dot points), B. Follow-up email (under 200 words), C. SMS (under 160 characters)*

What to check

- Has the AI removed **all** references to other students?
- Are there any promises of specific outcomes?
- Is the tone empathetic without being defensive?

Exercise A2: What You Learned

The comms safety checklist

Before sending any incident-related communication, check:

- **No other students identified** (directly or by implication)
- **No outcomes promised** until decisions are formally made
- **No evidence claimed** unless verified and available
- **No blame assigned** in either direction
- **Next steps are specific** (who to contact, by when, what happens next)
- **Tone is calm and factual** (not defensive, not dismissive)

The principle

> *AI drafts fast. Fast drafts are dangerous. The **human review step** is not optional – it is where governance happens.*

Try this at work

- Run every sensitive email through AI with the instruction "check this for privacy breaches, over-promises, and unverified claims"

Exercise A3

Attendance Triage Under Uncertainty

Working with Imperfect Data

The theory: real exports are never clean

The attendance CSV in the Messy Mini-Pack has deliberate problems:

- **Duplicates** – S1188 appears twice for the same date
- **Inconsistent codes** – "Unexplained", "LATE", "X" instead of standard codes
- **Missing fields** – blank YearLevel, blank AttendanceCode
- **Impossible values** – MinutesLate of -3 (negative time)
- **Typos and format issues** – "S12O2" (letter O), DD/MM/YYYY mixed with YYYY-MM-DD, non-existent homeroom "8Z"

What this exercise demonstrates

- AI spots data quality issues **before** you start analysis
- AI handles the tedious work of finding inconsistencies – you make the calls

Exercise A3: Attendance Triage

Instructions (20 minutes)

Paste the **Messy Attendance CSV** (Artifact 01 from Messy Mini-Pack), then add:

> *This attendance export has data quality issues. Please:*
> *1. List every data quality problem you find (duplicates, inconsistent codes, missing fields, impossible values, typos, format issues)*
> *2. For each problem, suggest how to fix it or what to verify*
> *3. After noting the issues, produce a triage list: students who need parent contact this week, ranked by urgency (highest absence/late pattern first)*
> *4. For each student on the triage list, draft a one-sentence contact note appropriate to their situation*
>
> *Where data is uncertain, note what you are assuming and what needs to be verified in the source system.*

What to look for

- Does AI catch the **S12O2 typo** (letter O instead of zero)?
- Does it flag the **negative minutes late** value?
- Does it identify the **duplicate S1188 row**?

Exercise A3: What You Learned

Data quality issues found (1 of 2)

Issue	Detail
Duplicate row	S1188 appears twice on 2026-03-04
Code: "Unexplained"	S1019 – should be "U"
Code: "LATE"	S1007 – should be "L"
Unknown code "X"	S1089, 2026-03-05
Impossible value	S1102 – MinutesLate = -3

Exercise A3: What You Learned (continued)

Data quality issues found (2 of 2)

Issue	Detail
Missing YearLevel	S1095, 2026-03-04
StudentID typo	"S12O2" – letter O instead of zero
Date format	S1048 uses DD/MM/YYYY instead of YYYY-MM-DD
Invalid homeroom	S1048 assigned "8Z" (does not exist)
Missing AttendanceCode	S1121, 2026-03-05

The principle

> *Clean data before you analyse. AI finds the mess. **You** decide the fixes.*

Exercise A4

Policy Alignment Change Log

AI as Compliance Translator

The theory: old policy vs new requirements

- Schools receive new regulatory bulletins regularly
- Comparing a local procedure against a new bulletin **by hand** takes hours
- AI produces a structured comparison in seconds – flagging gaps and required updates

The "change log" pattern

Ask AI to produce a table with columns:

Requirement → **Current procedure** → **Gap or risk** → **Required update** → **Owner** → **Evidence**

This pattern works for **any** regulatory comparison – not just incident recording.

Exercise A4: Policy Alignment

Instructions (15 minutes)

Paste **Artifact 09** (Old Local Procedure) AND **Artifact 10** (Authority Bulletin), then add:

> *Compare the school's local procedure with the authority bulletin. Output a change log table with columns: Requirement, CurrentProcedure, Gap/Risk, RequiredUpdate, Owner, Evidence. Then draft a revised procedure that meets the new minimum expectations.*

Key gaps to expect

- **Recording timeframe:** "within a few days" vs "within 24 hours"
- **Complaints acknowledgement:** "5 business days" vs good practice of 2
- **Unconfirmed marking:** Authority requires it; local procedure does not
- **Privacy guidance:** "provide enough detail" vs "do not disclose other students"

Exercise A4: What You Learned

Key takeaways

- AI performs **structured document comparison** faster and more thoroughly than manual review
- The change log format makes gaps visible and actionable – not buried in paragraph text
- AI can draft a revised procedure, but the **owner assignments and timeframes** need human input

The principle

> *AI translates between "what the authority requires" and "what your school currently does" – the gap analysis is instant, the implementation is human*

Try this at work

- Every time a new circular, bulletin, or regulatory update arrives, paste it alongside your current procedure
- Ask for a change log table – this becomes your compliance evidence
- Assign owners in a staff meeting, not in the AI output

Exercise A5

Messy Minutes to Action Register

Imposing Structure on Chaos

The theory: meeting notes are always messy

The messy exec meeting notes (Artifact 11) have deliberate problems:

- **Unclear ownership** – "send list to coordinators" (who? which ones?)
- **Vague dates** – "maybe Wed?", "due Tue (or Wed?)", "start soon"
- **Missing decisions** – "decision-ish" (was it actually decided?)
- **Ambiguous actions** – "need a plan" (what plan? who writes it?)

What this exercise demonstrates

- AI extracts action items even from terrible notes and **flags ambiguity**
- The output is a clean action register – you confirm owners, dates, and decisions

Exercise A5: Messy Minutes

Instructions (15 minutes)

Paste **Artifact 11** (Messy Exec Meeting Notes), then add:

> *These meeting notes are messy and incomplete. Please:*
> *1. Format as professional minutes with clear headings*
> *2. Extract action items into a table: Action, Owner, DueDate, Status*
> *3. Where owner/date is unclear, write "TBC" and suggest a reasonable value*
> *4. List items that sound like decisions but are not confirmed – mark "DECISION TBC"*
> *5. Draft a 5-sentence email summary asking attendees to confirm their action items*

What to look for

- Does AI flag "maybe Wed?" as uncertain?
- Does it handle "decision-ish" appropriately?
- Does the email ask attendees to **confirm**, not just inform?

Exercise A5: What You Learned

Key ambiguities in the messy notes

Item	Problem	Resolution needed
"Send list to coordinators"	Who sends? Which coordinators?	Owner and recipient TBC
"Due Tue (or Wed?)"	Conflicting dates	Confirm with Wellbeing Leader
"Decision-ish: increase supervision"	Was this formally decided?	Confirm with Principal
"Trial weekly digest – start soon"	No start date, no owner	Assign owner and date
"10-slide deck outline – who drafts?"	No owner assigned	Assign at next meeting

The principle

> *AI imposes **structure** on messy inputs. You impose **accountability** – confirming owners, dates, and decisions. Neither step works without the other.*

The Advanced Toolkit

Five patterns for messy data

Pattern	When to use	Key output
Known / Unconfirmed / Unknown	Conflicting sources	Fact classification list
Comms Safety Check	Any sensitive draft	Privacy and promise audit
Data Quality Audit	Messy exports / CSVs	Issue list with fix suggestions
Change Log Comparison	Old vs new requirements	Gap table with owners
Ambiguity Flagging	Vague meeting notes	Action register with TBCs

The golden rule for all advanced exercises

> *AI is fast. Fast is dangerous when the data is messy.* ***The human verification step is where safety happens.***

Bringing It Together

The AI Sandwich in practice

Raw input → **AI transforms** → **You verify** → **Clean output**

Your verification checklist

Before sending any AI output, check:

- **Facts** confirmed by 2+ sources?
- **People** identified by role or ID only (no names in general text)?
- **Promises** limited to what has been formally decided?
- **Evidence** claims backed by actual records?
- **Owners and dates** specific (not "someone" or "soon")?

Thank You

Tracy Anthony and **Dr Dennis Wollersheim**
Real Minds AI

hello@realmindsai.com.au | realmindsai.com.au

wisdom, amplified